

Automated Excel Sales Report

A practical Python automation project for transforming raw Excel sales data into a structured, business-ready report.

Project Type	Python / Excel Automation
Primary Tools	Pandas + OpenPyXL
Input	Raw Excel sales dataset
Output	Automated multi-sheet Excel report
Dataset Size	9,994 records × 21 columns

1. Business Problem

Businesses often receive sales data in Excel and repeatedly need the same checks, calculations, summaries, and formatting. Performing this process manually is repetitive and becomes inefficient when reports must be generated regularly.

The objective of this project was to replace that repetitive workflow with a reusable Python automation pipeline.

2. Solution

The automation combines Pandas for data analysis with OpenPyXL for Excel workbook manipulation and presentation. Pandas handles the analytical work, while OpenPyXL controls workbook structure and formatting.

3. Automated Workflow

Step	Automated Operation
1	Load and inspect the raw Excel dataset
2	Check duplicates and missing values
3	Validate negative Quantity, Sales, and Profit values
4	Calculate overall sales, profit, discount, and record metrics
5	Generate Category-level sales and profit summaries
6	Generate Region-level sales and profit summaries
7	Export summaries to separate Excel sheets
8	Create a Report Summary sheet
9	Apply professional formatting with OpenPyXL
10	Save a reusable final Excel report

4. Dataset & Data Quality

The working dataset contains 9,994 records and 21 columns, including order information, customers, locations, product categories, sales, quantity, discount, and profit.

Check	Result
Duplicate rows	0
Missing values	0 across all columns
Negative Quantity	0
Negative Sales	0
Negative Profit	1,871 transactions

5. Business Analysis

The project calculates overall business metrics and creates grouped summaries for management-style reporting.

Analysis	Purpose
Total Sales	Overall revenue generated in the dataset
Total Profit	Overall profit across transactions
Average Discount	Average discount applied
Category Summary	Compare sales and profit by product category
Region Summary	Compare sales and profit by region

6. Final Excel Report

The generated workbook contains three main sheets. The report is designed to be readable by a business user without requiring them to inspect the Python code.

Sheet	Purpose
Report Summary	High-level KPIs and data-quality indicators
Category Summary	Sales and profit grouped by category
Region Summary	Sales and profit grouped by region

7. Excel Automation Features

OpenPyXL is used to turn the analytical output into a reusable Excel deliverable. The automation creates and updates worksheets, writes values, styles headers, formats numbers, adjusts column widths, merges the report title, and controls workbook presentation.

- Multiple worksheet creation and management
- Automated Excel export using Pandas ExcelWriter
- Bold and filled report headers
- Number formatting using #,##0.00
- Percentage formatting
- Automatic column-width calculation where appropriate
- Merged report title
- Freeze panes for tabular report sheets
- Re-runnable report generation without duplicate Report Summary sheets

8. Technologies

Python — automation language

Pandas — data loading, validation, aggregation, and analysis

OpenPyXL — workbook manipulation and Excel formatting

Excel — final business-facing report format

9. Project Structure

```
automated-excel-sales-report/  
■■■ data/  
■ ■■■ Sample-sales-data-excel.xlsx  
■■■ output/  
■ ■■■ Sales_Report.xlsx  
■■■ src/  
■ ■■■ sales_report.py  
■■■ docs/  
■ ■■■ project_report.pdf  
■■■ README.md
```

10. Skills Demonstrated

Python automation • Excel automation • Pandas data analysis • OpenPyXL • data validation • business reporting • report generation • reusable scripts • structured project organization

11. Portfolio Value

This project demonstrates a complete business-oriented automation workflow rather than an isolated Python exercise. It shows how raw business data can be transformed into a repeatable, client-facing report. The same pattern can be adapted to recurring sales, financial, inventory, operational, and other Excel-based reporting tasks.

Project documentation prepared for portfolio, GitHub, and professional presentation.